

Authorship

UA CITI Ethics Training

The Importance of Authorship

Authorship is the way that the research community identifies those who should be both recognized and credited with a specific contribution to the work and held responsible and accountable for the information contained in publications. Authorship is an important measure of intellectual activity and achievement.

Receiving credit for one's contribution in the form of authorship is usually critical for professional advancement because it is a primary basis for hiring, for the awarding of grants and scholarships, and for promotion and tenure.



Giving Credit

- An ethical writer ALWAYS acknowledges the contributions of others to his/her work.
- In a journal paper, in what ways can you give credit?
 - Co-authorship
 - Acknowledgments Section
 - Citations & “Private Communications”

Giving Credit

- An ethical writer ALWAYS acknowledges the contributions of others to his/her work.
- In a journal paper, in what ways can you give credit?
 - Authorship
 - Ordering of Authorship
 - Acknowledgements Section
 - Software Section
 - References in the paper itself.
 - Cite as “Private communication, 20XX”

Who should be an author?

- From Homework: What were the reasons for the authors (and ordering) for the paper you picked?

Who should be an author?

- From Homework: What were the reasons for the authors (and ordering) for the paper you picked?

Manikantan + 2024

- Supervisor (last)
- 2 direct involved
- Conflict resolution/solution
- ?
- Data/simulations. → Architect

Laliotis + 2024

- Supervisor (third)
- 2nd author – initial analysis
- 4th – restructuring of code
- 5th – on precursor paper → Proposals
- 6th – simulations – Architect

Meyer et al + 2026

- last author – Sanity Check -→ Appeal to Authority

Franz et al

- large author list, alphabetical (twice) – large group invited to be authors after the paper is pretty much done.

Who should be an author?

Each author is responsible for the integrity of their own work and should have confidence in the integrity of the manuscript as a whole.

<https://planetary-research.org/authorship>

CITI Training

Among the most widely discussed current authorship standards are those provided by the International Committee of Medical Journal Editors (ICMJE 2017). This organization recommends that authorship be based on these criteria:

Substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work;

Drafting the work or revising it critically for important intellectual content;

Final approval of the version to be published; and

Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Contributor Role Taxonomy (CRediT)

<https://credit.niso.org/>

NISO : *National Information Standards Organization*, a nonprofit membership organization that identifies, develops, maintains, and publishes technical standards to manage information

Exercise: Look at the list of authors from your homework

<https://planetary-research.org/authorship>

Role	Description
Conceptualization	Ideas; formulation or evolution of overarching research goals and aims.
Data curation	Management activities to annotate (produce metadata), scrub data and maintain research data (including software code, where it is necessary for interpreting the data itself) for initial use and later re-use.
Formal analysis	Application of statistical, mathematical, computational, or other formal techniques to analyze or synthesize study data.
Funding acquisition	Acquisition of the financial support for the project leading to this publication.
Investigation	Conducting a research and investigation process, specifically performing the experiments, or data/evidence collection.
Methodology	Development or design of methodology; creation of models.
Project administration	Management and coordination responsibility for the research activity planning and execution.
Resources	Provision of study materials, reagents, materials, patients, laboratory samples, animals, instrumentation, computing resources, or other analysis tools.
Software	Programming, software development; designing computer programs; implementation of the computer code and supporting algorithms; testing of existing code components.
Supervision	Oversight and leadership responsibility for the research activity planning and execution, including mentorship external to the core team.
Validation	Verification, whether as a part of the activity or separate, of the overall replication/reproducibility of results/experiments and other research outputs.
Visualization	Preparation, creation and/or presentation of the published work, specifically visualization/data presentation.
Writing – original draft	Preparation, creation and/or presentation of the published work, specifically writing the initial draft (including substantive translation).
Writing – review & editing	Preparation, creation and/or presentation of the published work by those from the original research group, specifically critical review, commentary or revision – including pre- or post-publication stages.

Note About Errors & Co-Author Responsibility

-- Errata

- <https://journals.aas.org/manuscript-preparation/#errata>

It's ok to make honest mistakes.

Errata are intended to correct errors in the original articles and should not contain updates or other new results.

They should also not be written for insignificant or minor corrections such as author affiliation updates or author ordering changes. Co-authors can help you figure out if an errata is warranted.

In all cases, the responsible Scientific Editor must approve each Erratum before it can be published, and may require that the Erratum be sent for peer review by the original referee or by an independent referee.

Awarding Credit: Plagiarism of Ideas

CREDIT WHERE CREDIT IS DUE

Ben, a third-year graduate student, had been working on a research project that involved an important new experimental technique. For a national meeting in his discipline, Ben wrote an abstract and gave a brief presentation that mentioned the new technique. After his presentation, he was surprised and pleased when Dr. Freeman, a leading researcher from another university, engaged him in an extended conversation. Dr. Freeman asked Ben extensively about the new technique, and Ben described it fully. Ben's own faculty advisor often encouraged his students not to keep secrets from other researchers, and Ben was flattered that Dr. Freeman would be so interested in his work.

Six months later Ben was leafing through a journal when he noticed an article by Dr. Freeman. The article described an experiment that clearly depended on the technique that Ben had developed. He didn't mind ; in fact, he was again somewhat flattered that his technique had so strongly influenced Dr Freeman's work. But when he turned to the citations, expecting to see a reference to his abstract or presentation, his name was nowhere to be found.

1. Does Ben have any way of receiving credit for his work?
2. Should he contact Dr. Freeman in an effort to have his work recognized?
3. Is Ben's faculty advisor mistaken in encouraging his students to be so open about their work?

On Being a Scientist: Responsible Conduct in Research 2000, The National Academy of Sciences

Acknowledgement vs. a reference in the paper?

- How do you decide?

Authorship vs. Acknowledgement vs. Reference in the paper

- References serve to acknowledge published (or soon to be published) work in the field
- Acknowledgements section in paper for:
 - Conversations that supported the work that are not already published.
 - Acknowledging Code/Tools that enabled you to do the science or write the paper.
 - Funding Sources
 - Other programs that supported the work (e.g. TAP!).

Acknowledgements are searchable in ADS and are often the way funders keep track of impact of their programs.

References – How do you know if you’ve given everyone proper credit?

- My Suggestions:
- Post to arxiv before paper is accepted (e.g. after first ref report) and add “comments welcome”
- Before the paper is accepted by the journal, send a copy of the paper (arxiv link) to the folks you have cited and ask them if you have properly cited their work.
 - People will look for their name.
 - You still have time to make changes

APS Statement: Senior Authors

Supervisors and/or other senior scientists should not be listed on papers of subordinates unless they have also contributed significantly to the concept, design, execution or interpretation of the research study.

BUT "appeals to authority" can be powerful support for junior authors.

Awarding Credit

WHO SHOULD GET CREDIT FOR THE DISCOVERY OF PULSARS?

A much-discussed example of the difficulties associated with allocating credit between junior and senior researchers was the 1967 discovery by Jocelyn Bell,

then a 24-year-old graduate student, of pulsars. Over the previous two years, Bell and several other students, under the supervision of Bell's thesis advisor, Anthony Hewish, had built a 4.5-acre radiotelescope to investigate scintillating radio sources in the sky. After the telescope began functioning, Bell was in charge of operating it and analyzing its data under Hewish's direction. One day Bell noticed "a bit of scruff" on the data chart. She remembered seeing the same signal earlier and by measuring the period of its recurrence, determined that it had to be coming from an extraterrestrial source.

Together Bell and Hewish analyzed the signal and found several similar examples elsewhere in the sky. After discarding the idea that the signals were coming from an extraterrestrial intelligence, Hewish, Bell, and three other people involved in the project published a paper announcing the discovery, which was given the name "pulsar" by a British science reporter.

Many argued that Bell should have shared the Nobel Prize awarded to Hewish for the discovery, saying that her recognition of the signal was the crucial act of discovery. Others, including Bell herself, said that she had received adequate recognition in other ways and should not have been so lavishly rewarded for doing what a graduate student is expected to do in a project conceived and set up by others.

On Being a Scientist: Responsible Conduct in Research 2000, The National Academy of Sciences

Hewish, Bell et al. 1968

Interesting Series about Jocelyn Bell Burnell and the discovery of pulsars:

PART 4:

https://www.youtube.com/watch?v=zsi_TJSo-A

Authorship Ordering: Why Clarity is Needed

In the absence of a clear and consistent policy, members of the community often attempt to infer something about author contributions from the ordering of the author list, such as when reviewing applications for jobs, promotion and tenure packets, or admissions to PhD programs.

<https://journals.aas.org/author-contribution-statements/>

Authorship Ordering: Why Clarity is needed

- BUT authorship ordering is not standardized in Astro/Planetary Sci
 - authors can be listed in order of effort expended on the project
 - students and early career researchers can be listed first for increased visibility.
 - Some large collaborations specify that authorship in articles must be listed alphabetically
 - authors can be listed by tiers according to their level of contribution, and may or may not list them alphabetically within each tier.
 - other collaborations have **specific agreements** in place that prescribe authorship criteria and listed author order.
 - Furthermore, as names must be listed sequentially in some way, co-leadership of a paper cannot easily be indicated.

<https://journals.aas.org/author-contribution-statements/>

Authorship Contribution Statements

- <https://journals.aas.org/author-contribution-statements/>
- AAS Journals (ApJ, PSJ) now offer an “Authorship Contribution” section
- Author Contributions provide an opportunity for increased transparency and ensure that everyone receives appropriate credit for their work.
- MNRAS says you can do it in the Acknowledgements section

account for the full interaction history of the system. In this way, the halo-disk connection becomes more concrete in the presence of a satellite, providing an exciting path forwards to constrain the nature of DM.

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We respectfully acknowledge the University of Arizona is on the land and territories of Indigenous peoples. Today, Arizona is home to 22 federally recognized tribes, with Tucson being home to the O’odham and the Yaqui. Committed to diversity and inclusion, the University strives to build sustainable relationships with sovereign Native Nations and Indigenous communities through education offerings, partnerships, and community service.

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AUTHOR CONTRIBUTIONS

HRF was responsible for the BFE construction and analysis, and wrote and submitted the manuscript. HR developed and ran the N -body simulation, contributed Section 2.1, and edited the manuscript. GB supervised HRF and HR, obtained funding, provided guidance on the analysis, and edited the manuscript. NGC edited the manuscript, and provided guidance on the analysis as well as a snapshot of his simulation described in N. Garavito-Camargo et al. (2019). EP contributed the rigid-potential orbits in Section 3.4 and edited the manuscript. MSP and MDW provided guidance on the analysis and edited the manuscript. FAG and CFPL provided significant comments on an early draft of the manuscript and edited the final version.

Software: **Astropy** (Astropy Collaboration et al. 2013, 2018, 2022); **EXP/pyEXP** (M. S. Petersen & M. D. Weinberg 2025); **Gadget-4** (V. Springel et al. 2021); **H5py** (A. Collette et al. 2020); **Healpy** (K. M. Górski et al. 2005; A. Zonca et al. 2019); **Jupyter** (T. Kluyver et al. 2016; B. E. Granger & F. Pérez 2021); **Matplotlib** (J. D. Hunter 2007); **Numpy** (C. R. Harris et al. 2020); **Screenshot** (en et al. 2020);

[From Foote + 2026](#)

Ack:
Reviewer
Discussions
Grants
Land Ack
TAP
HPC
NASA ADS
ArXiv